Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No:92301733021**

Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression

IDE: Google Colab

Theory:

Linear regression is a method for determining the best linear relationship between two variables *X* and *Y*. If variables *X* and *Y* are uncorrelated, it is pointless embarking upon linear regression. However, if a reasonable degree of correlation exists between *X* and *Y* then linear regression may be a useful means to describe the relationship between the two variables. The usual approach is to use the *least-squares* method, which minimizes the squared difference between the actual data points and a straight line. Let $[x_i, y_i]$, $i = 1,2,3,\ldots,N$ be the *N* pairs of data values of the variables *X* and *Y*. The straight-line relating *X* and *Y* is $y = mx + c$, where *m* and *c* are the gradient and constant values (to be determined) defining the straight line. Thus, $y(x_i) - y_i$ is the difference between the line and data point *i* (see Fig. 1). Taking all the data points, we seek values of *m* and *c* that minimize the squared difference *SD*.

$$\sum_{1}^{N} \left[y(x_i) - y_i \right]^2$$

This is achieved by calculating the partial derivatives of *SD* with respect to *m* and *c* and finding the pair [*m*,*c*] such that *SD* is at a minimum.

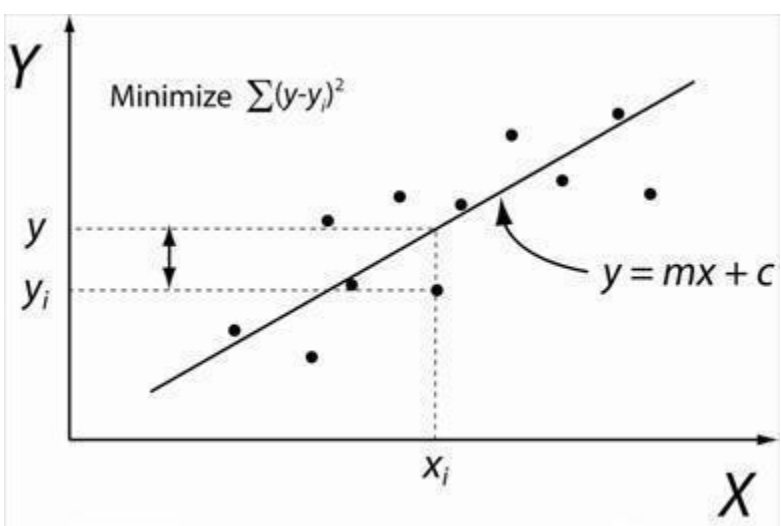

Figure 1: Illustration of Linear Regression. Linear least squares regression, the idea is to find the line y = mx + c that minimizes the mean squared difference between the line and the data points

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Batch Gradient Descent:

Gradient Descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms. It is basically used for updating the parameters of the learning model. Batch gradient descent which processes all the training examples for each iteration of gradient descent. But if the number of training examples is large, then batch gradient descent is computationally very expensive.

Let m be the number of training examples. Let n be the number of features.

Algorithm for batch gradient descent :

Let $h_\theta(x)$ be the hypothesis for linear regression. Then, the cost function is given by:
Let Σ represents the sum of all training examples from i=1 to m.
$J_{train}(\theta) = (1/2m)\ \Sigma(\ h_\theta(x^{(i)})\ - y^{(i)})^2$

Repeat {
$\theta j = \theta j - (\text{learning rate}/m) * \Sigma(\ h_\theta(x^{(i)})\ - y^{(i)})x_j^{(i)}$
For every j =0 …n
}
Where $x_j^{(i)}$ Represents the jth feature of the ith training example. So if *m* is very large(e.g. 5 million training samples), then it takes hours or even days to converge to the global minimum.That's why for large datasets, it is not recommended to use batch gradient descent as it slows down the learning.

Pre Lab Exercise:

a. Explain the meaning of linear regression

Linear regression is a machine learning technique used to predict a continuous value by finding a straight-line relationship between input variables and output. It tries to fit the best line that represents the data.

b. Write three applications of linear regression

It is used in predicting house prices, estimating sales based on advertising, and forecasting trends like temperature or stock prices. These applications rely on finding relationships between variables.

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c. Write three advantages of linear regression

Linear regression is simple to understand and easy to implement. It works well with small datasets and provides clear insights into relationships between variables.

d. Write three limitations of linear regression

It assumes a linear relationship, which may not always be true. It is sensitive to outliers and cannot handle complex patterns or non-linear data effectively.

Methodology:

1. Load the basic libraries and packages

Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

2. Load the dataset

Code:

```
url = "https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv"
df = pd.read_csv(url)
```

3. Analyse the dataset

Code:

```
print("First 5 rows:\n", df.head())
print("\nDataset Info:\n")
print(df.info())
print("\nStatistics:\n", df.describe())
```

Output:

```
First 5 rows:
    Unnamed: 0     TV  radio  newspaper  sales
0           1  230.1   37.8       69.2   22.1
1           2   44.5   39.3       45.1   10.4
2           3   17.2   45.9       69.3    9.3
3           4  151.5   41.3       58.5   18.5
4           5  180.8   10.8       58.4   12.9

Dataset Info:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Unnamed: 0  200 non-null    int64
 1   TV          200 non-null    float64
 2   radio       200 non-null    float64
 3   newspaper   200 non-null    float64
 4   sales       200 non-null    float64
dtypes: float64(4), int64(1)
memory usage: 7.9 KB
None

Statistics:
        Unnamed: 0          TV       radio   newspaper       sales
count  200.000000  200.000000  200.000000  200.000000  200.000000
mean   100.500000  147.042500   23.264000   30.554000   14.022500
std     57.879185   85.854236   14.846809   21.778621    5.217457
min      1.000000    0.700000    0.000000    0.300000    1.600000
25%     50.750000   74.375000    9.975000   12.750000   10.375000
50%    100.500000  149.750000   22.900000   25.750000   12.900000
75%    150.250000  218.825000   36.525000   45.100000   17.400000
max    200.000000  296.400000   49.600000  114.000000   27.000000
```

4. Pre-process the data

Code:

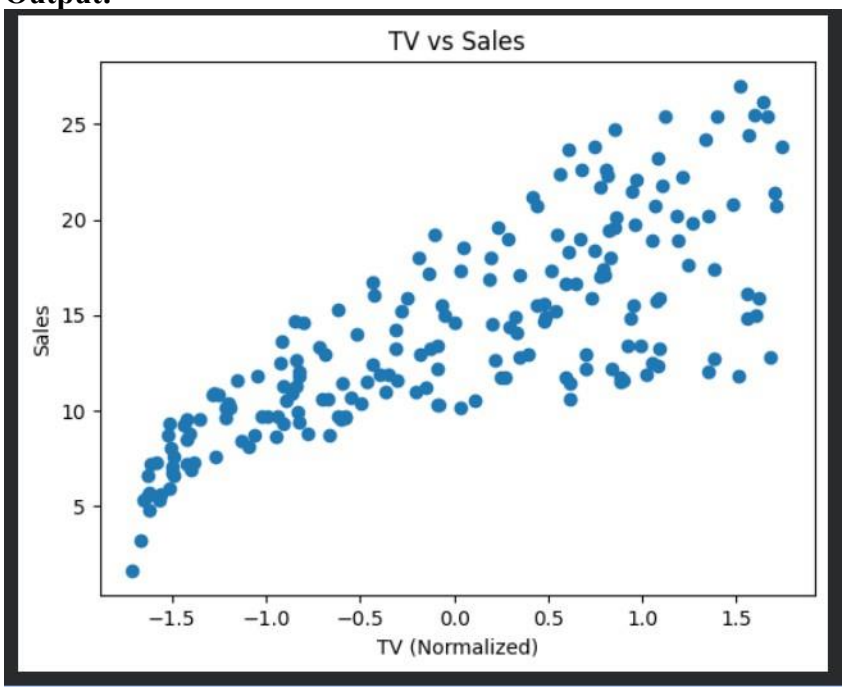
```
# Use only TV advertising for simple linear regression
X = df['TV'].values
y = df['sales'].values

# Normalize data (optional but good practice)
X = (X - X.mean()) / X.std()
```

5. Visualize the Data

Code:

```
plt.scatter(X, y)
plt.title("TV vs Sales")
plt.xlabel("TV (Normalized)")
plt.ylabel("Sales")
plt.show()
```

Output:



6. Separate the feature and prediction value columns

Code:

```
X = X.reshape(-1, 1)
m = len(y)

# Add bias term (column of 1s)
X_b = np.c_[np.ones((m, 1)), X]
```

7. Write the Hypothesis Function

Code:

```
def hypothesis(X, theta):
    return np.dot(X, theta)
```

8. Write the Cost Function

Code:

```
def cost_function(X, y, theta):
    m = len(y)
    return (1/(2*m)) * np.sum((hypothesis(X, theta) - y) ** 2)
```

9. Write the Gradient Descent optimization algorithm

Code:

```
def gradient_descent(X, y, theta, lr, iterations):
    m = len(y)
    cost_history = []

    for i in range(iterations):
        gradients = (1/m) * np.dot(X.T, (hypothesis(X, theta) - y))
        theta = theta - lr * gradients
        cost_history.append(cost_function(X, y, theta))

    return theta, cost_history
```

10. Apply the training over the dataset to minimize the loss

Code:

```
theta = np.zeros(2)  # initial parameters
learning_rate = 0.01
iterations = 1000

theta, cost_history = gradient_descent(X_b, y, theta, learning_rate, iterations)

print("\nBest parameters (theta):", theta)
```

Output:

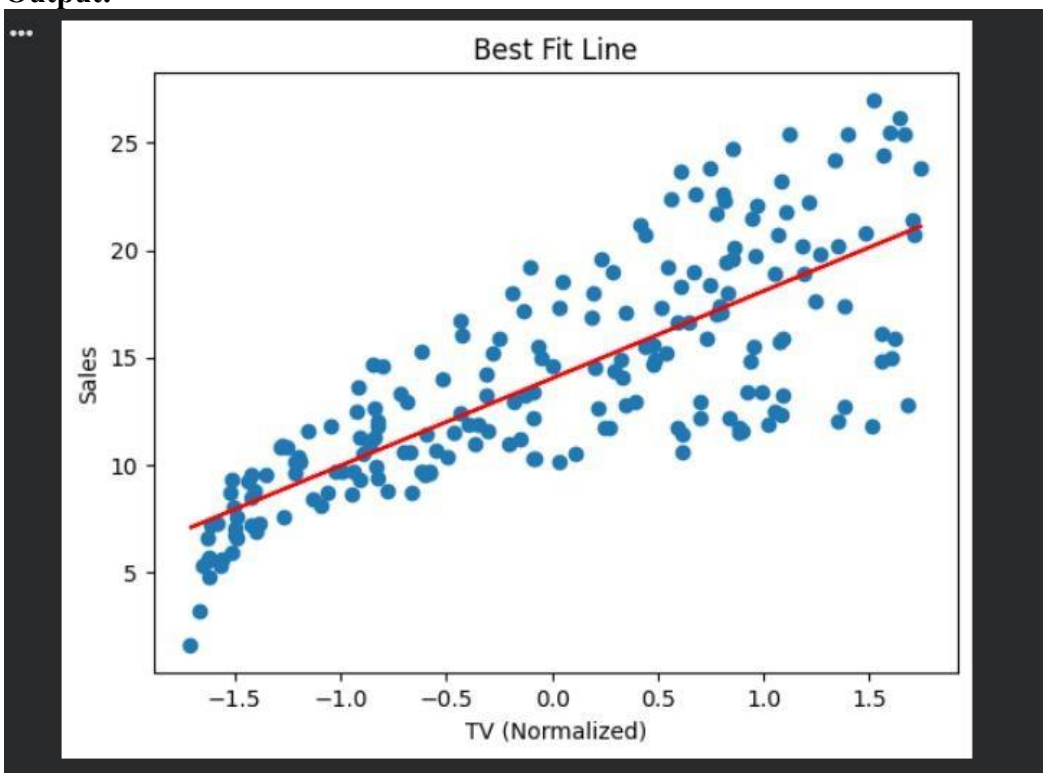
```
...
    Best parameters (theta): [     14.022      4.0708]
```

11. Find the best fit line to the given dataset

Code:

```
y_pred = hypothesis(X_b, theta)

plt.scatter(X, y)
plt.plot(X, y_pred, color='red')
plt.title("Best Fit Line")
plt.xlabel("TV (Normalized)")
plt.ylabel("Sales")
plt.show()
```

Output:

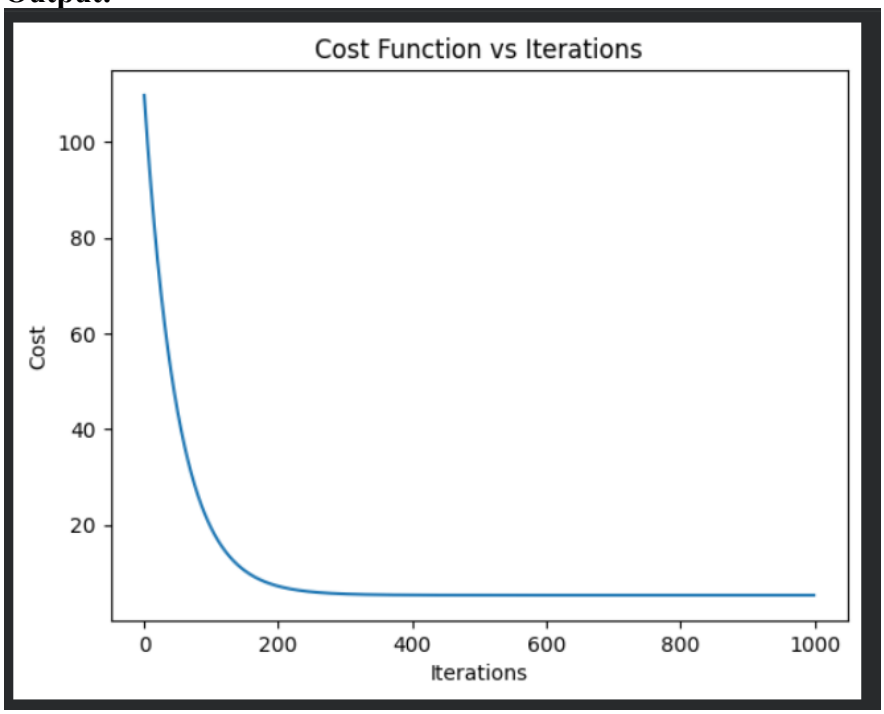


12. Observe the cost function vs iterations learning curve

Code:

```
plt.plot(range(iterations), cost_history)
plt.title("Cost Function vs Iterations")
plt.xlabel("Iterations")
plt.ylabel("Cost")
plt.show()
```

Output:



Program (Code):

To be attached with

Results:

To be attached with

a. Datapoints scattering (without best fit line)
b. Scatter plot showing the Best fit line in the last iteration of training

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Observation and Result Analysis:

a. Nature of the dataset

The dataset contains advertising data with features like TV, Radio, and Newspaper budgets, and the target variable is sales. In this code, only the **TV feature** is used, showing a generally linear relationship with sales. The data is numerical and suitable for simple linear regression.

b. During Training Process

During training, the model uses gradient descent to update the parameters (θ values) step by step. The cost function gradually decreases with each iteration, indicating that the model is learning and improving its predictions. The learning curve shows this reduction in error over time.

c. After the training Process

After training, the model finds the best-fit line that represents the relationship between TV advertising and sales. The predicted values closely follow the actual data points, showing a good fit. The final θ values represent the optimized parameters with minimum error.

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Post Lab Exercise:

a. What are the major assumptions considered in linear regression

Linear regression assumes a linear relationship between input and output variables. It also assumes that errors are normally distributed, independent, and have constant variance (homoscedasticity). Additionally, there should be no strong multicollinearity between features.

b. Why MSE is used instead of MAE for calculating the loss function

MSE is preferred because it squares the errors, giving more importance to large errors and making the model more sensitive to them. It is also mathematically easier to optimize using gradient descent since it is differentiable everywhere, unlike MAE.

c. How can the behaviour of outliers be understood while dealing with the unseen dataset

Outliers can be identified using techniques like IQR, boxplots, or Z-score. When dealing with unseen data, similar statistical checks are applied to detect extreme values. Proper preprocessing or robust models help reduce their impact.

d. Derive the Normal Equation for the Linear Regression.

The Normal Equation is used to directly find the optimal parameters without iteration.

$$\theta = (X^T X)^{-1} X^T y$$

It is derived by minimizing the cost function and setting its derivative to zero. This gives a closed-form solution for θ, which directly computes the best-fit parameters.